

SAT Math Hard Question 6

1

★ Mark for Review

ABC

Given the expression

$$\sqrt[5]{70n \left(\sqrt[7]{70n} \right)^2}$$

for what value of x is the given expression equivalent to $(70n)^{30x}$, where $n > 1$?

2

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ABC

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as

$$(hx + k)(x + j)$$

where h, k, j are integer constants. Which of the following must be an integer?

A. $\frac{b}{h}$

B. $\frac{b}{k}$

C. $\frac{45}{h}$

D. $\frac{45}{k}$

3

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ABC

The expression

$$6\sqrt[5]{3x^{45}} \cdot \sqrt[8]{8x}$$

is equivalent to ax^b , where a and b are positive constants and $x > 1$. What is the value of b ?

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ABC

In the given equation, c is a positive constant. Solve for x :

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

Which of the following is one of the solutions to the given equation?

A. $-c$

B. $-c^2 - 39^2$

C. $-\sqrt{39^2 - c^2}$

D. $\sqrt{c^2 + 39^2}$

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ABC

The quadratic function is given as:

$$f(x) = ax^2 + 4x + c$$

where a and c are constants. The graph of $y = f(x)$ is a parabola that opens upward and has a vertex at (h, k) , where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

I $c < 0$

II $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II

6

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ABC

The functions f and g are defined by the given equations, where $x \geq 0$:

I $f(x) = 33(0.4)^{x+3}$

II $g(x) = 33(0.16)(0.4)^{x-2}$

Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

A. I only

B. II only

C. I and II

D. Neither I nor II

7

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ABC

Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the **smallest standard deviation**?

Value	Frequency (A)	Frequency (B)	Frequency (C)	Frequency (D)
5	0	11	15	18
6	10	17	15	14
7	55	19	15	11
8	10	17	15	14
9	0	11	15	18

A. A

B. B

C. C

D. D

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ABC

In $\triangle ABC$, $AB = 4$ and $BC = 5$. If $\angle ABC < 90^\circ$, which of the following could be the area of $\triangle ABC$?

A. 5

B. 10

C. 15

D. 20

9

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ABC

The table summarizes the distribution of heights, in inches (“in”), for 300 participants in a study. If a participant that is at most 71 inches tall is chosen at random, what is the probability that the participant was selected from Group 2?

	48–59 in.	60–71 in.	72–83 in.	Total
Group 1	20	60	20	100
Group 2	10	75	15	100
Group 3	30	45	25	100
Total	60	180	60	300

A. $\frac{1}{4}$

B. $\frac{17}{48}$

C. $\frac{17}{60}$

D. $\frac{17}{20}$

10

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ABC

The function f is defined as $f(x) = x^2 + ax + b$, where a and b are positive integers. If the minimum value of $f(x)$ is 7, which is a possible value of b ?

A. 6

B. 7

C. 8

D. 9

11

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ABC

The average annual energy cost for a certain home is \$4,334. The homeowner plans to spend \$25,000 to install a geothermal heating system. The homeowner estimates that the average annual energy cost will then be \$2,712. Which of the following inequalities can be solved to find t , the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

A. $25,000 > (4,334 - 2,712)t$

B. $25,000 < (4,334 - 2,712)t$

C. $25,000 - 4,334 - 2,712t$

D. $25000 > \frac{4332}{2712}t$

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ABC

For the equation, b and c are constants. For any real number t , the point

$$\left(\frac{-2t + 4}{3}, t \right)$$

lies on the graph of the equation $bx + 3y = c$. What is the value of c ?

A. -2

B. 4

C. 6

D. 12

13

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ABC

A circle in the xy -plane has equation

$$(x - 2.5)^2 + (y - 3.5)^2 = 225.$$

Line p is tangent to the circle at point $(-6.5, C)$, where C is a constant. Which of the following could be the slope of line p ?

I. $-\frac{3}{4}$

II. $\frac{3}{4}$

III. $4/3$

A. I only

B. I and II only

C. I and III only

D. I, II, and III

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ABC

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the points $(c, 7)$ and $(2c, 117)$, where c is a constant. What is the value of b ?

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ABC

The given equation, where t is a positive constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{41}$. What is the value of t ?

$$2x^2 + 2y^2 + tx + ty - 78 = 0$$